

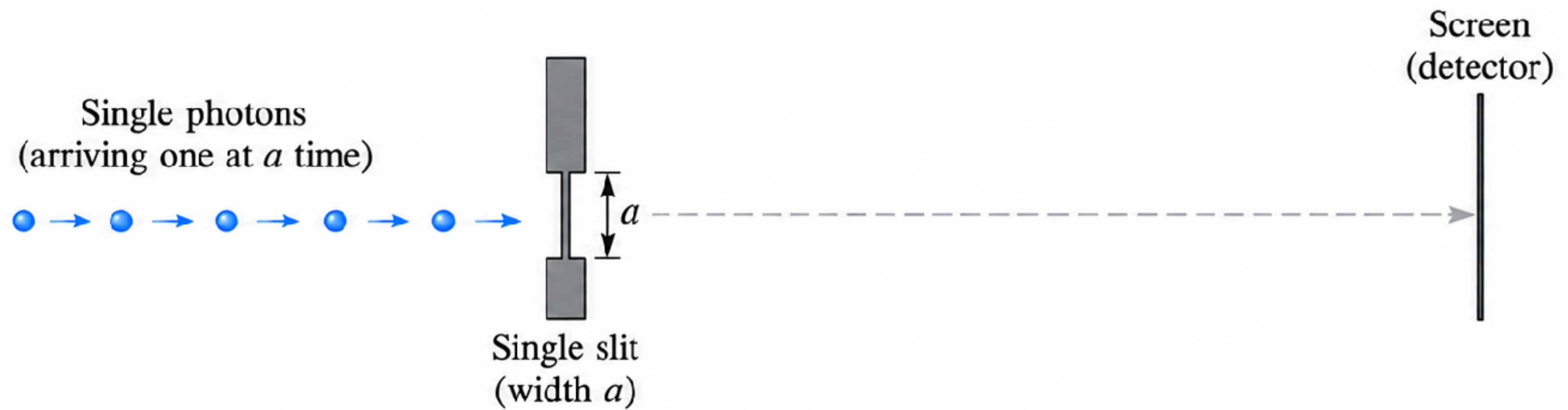
Modern Physics

Lecture 4: DeBroglie Wave and Uncertainty Principle

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Books: *Ch 3 of* Arthur Beiser,
Concepts of Modern Physics
Ch 38.4 of Hugh D. Young and Roger A.
Freedman, *University Physics with Modern Physics*

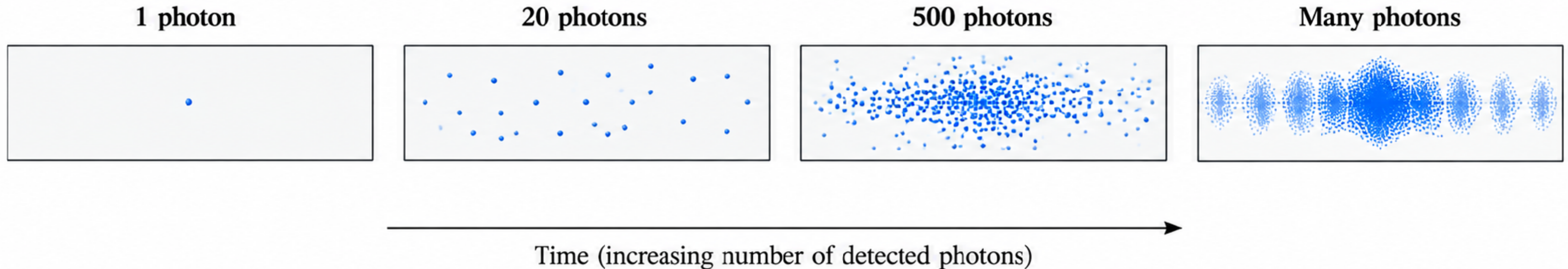
Recap: Diffraction in Photon picture



One particle makes one hit on screen. All Photons have same energy.

Recap: Diffraction in Photon picture

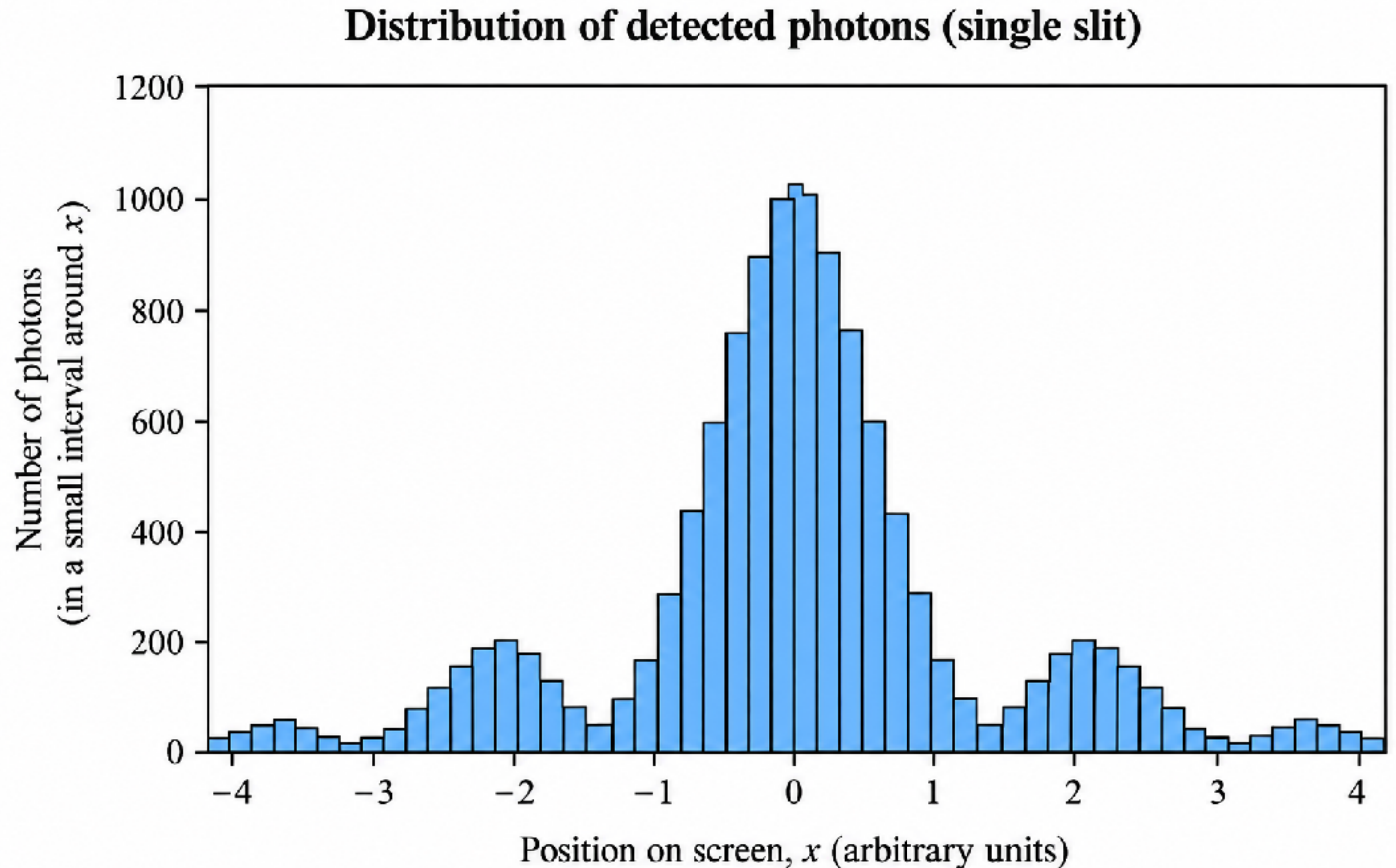
We repeat this 5000 times, keeping the location of every previous hit visible on the screen.



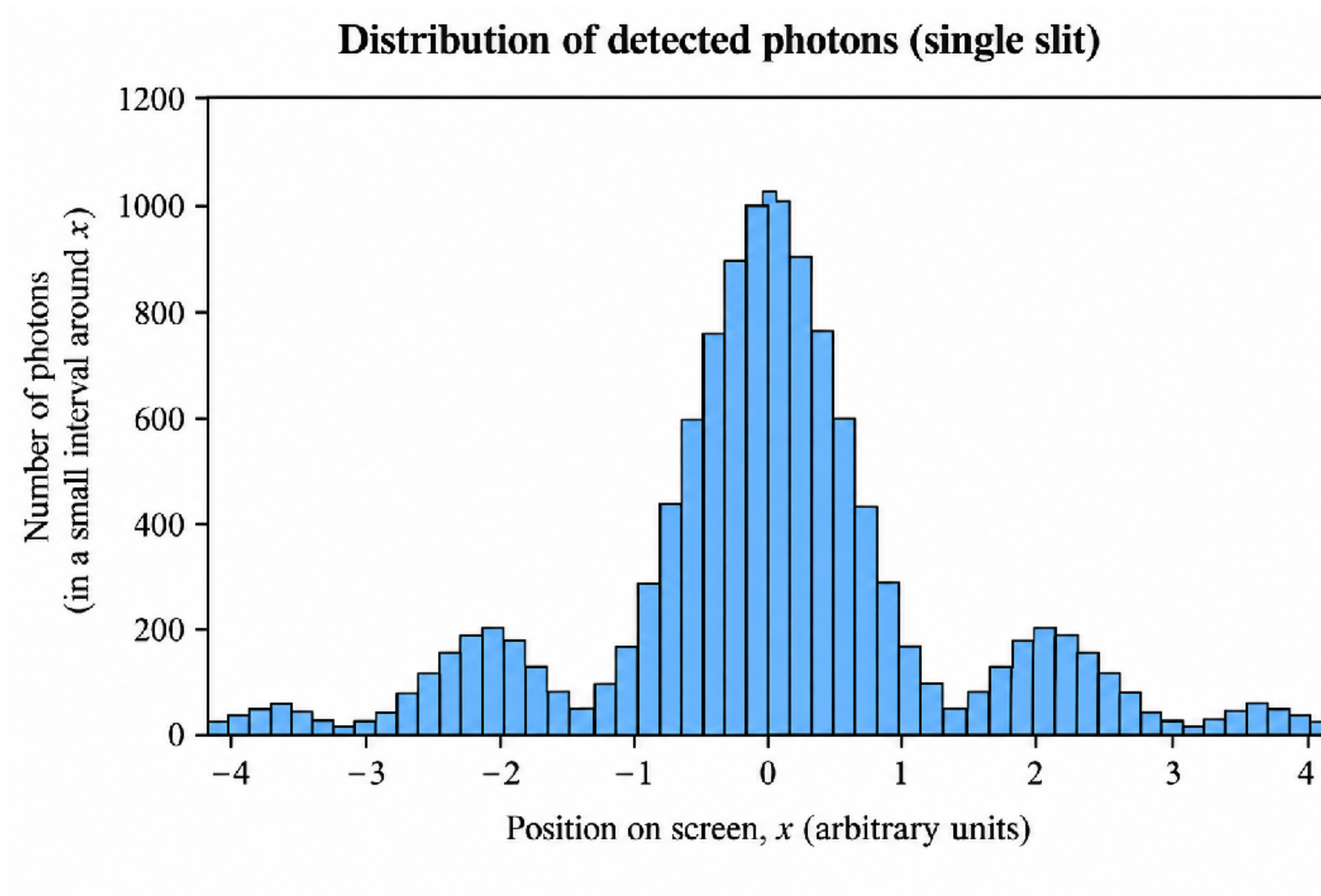
One photon $\times$ 500 identical experiments = 500 photons sent one at a time through one experiment.

Diffraction in Photon picture

- We cannot predict exactly where an individual photon will be detected, but different positions do not have equal probabilities.
- The collection of individual photon detections reproduces exactly the diffraction pattern predicted by wave optics.
- So the wave description has not disappeared when we introduce photons. It determines the statistical distribution of photon detections.



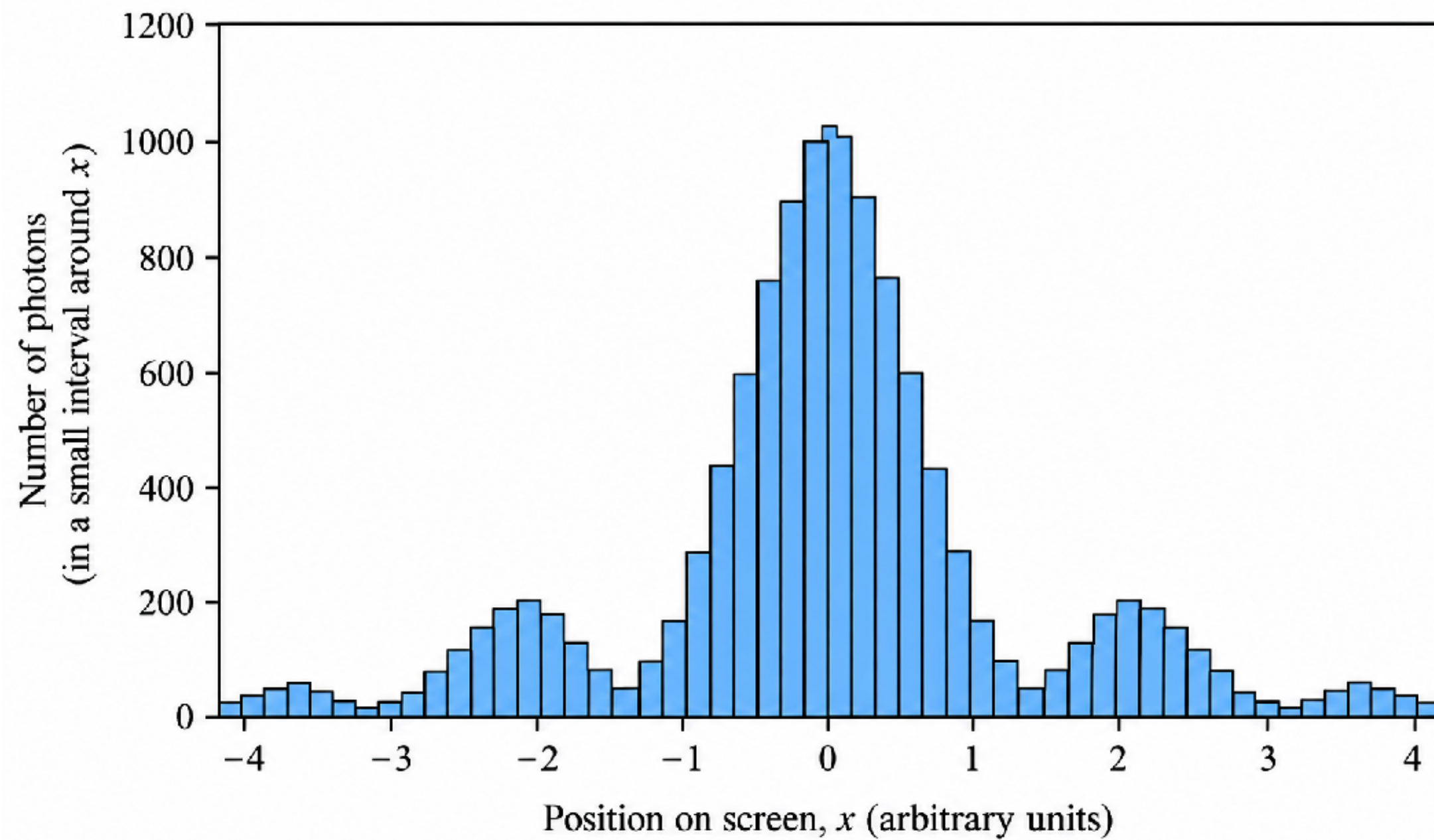
Diffraction in Photon picture



Identical to Diffraction Pattern

Diffraction in Photon picture

Distribution of detected photons (single slit)



Wave picture:

x: position

y: Intensity(y)

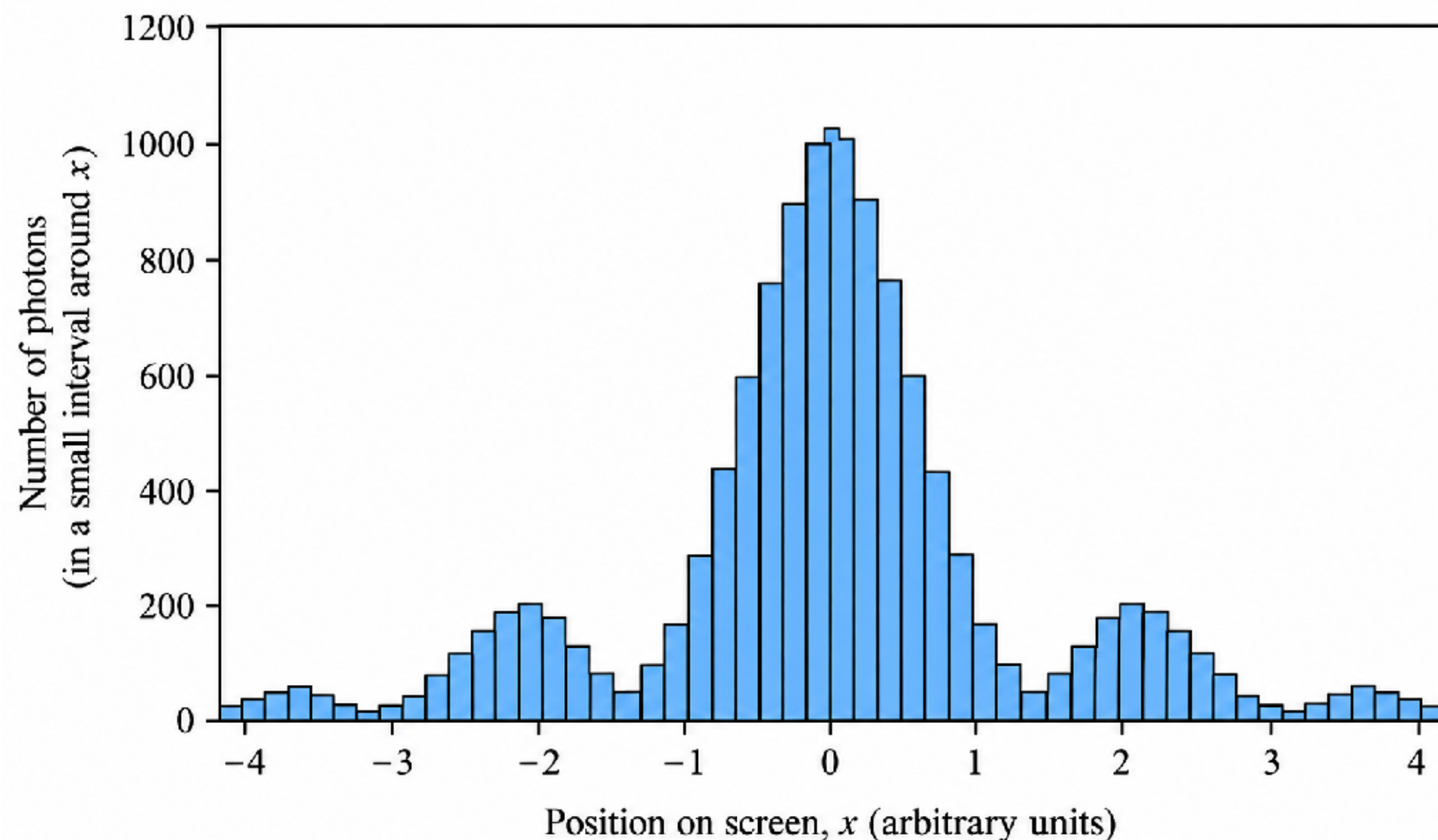
Particle picture:

x: position

y: $N(y)$

Diffraction in Photon picture

Distribution of detected photons (single slit)



Wave picture:

x: position

y: Intensity(y)

Particle picture:

x: position

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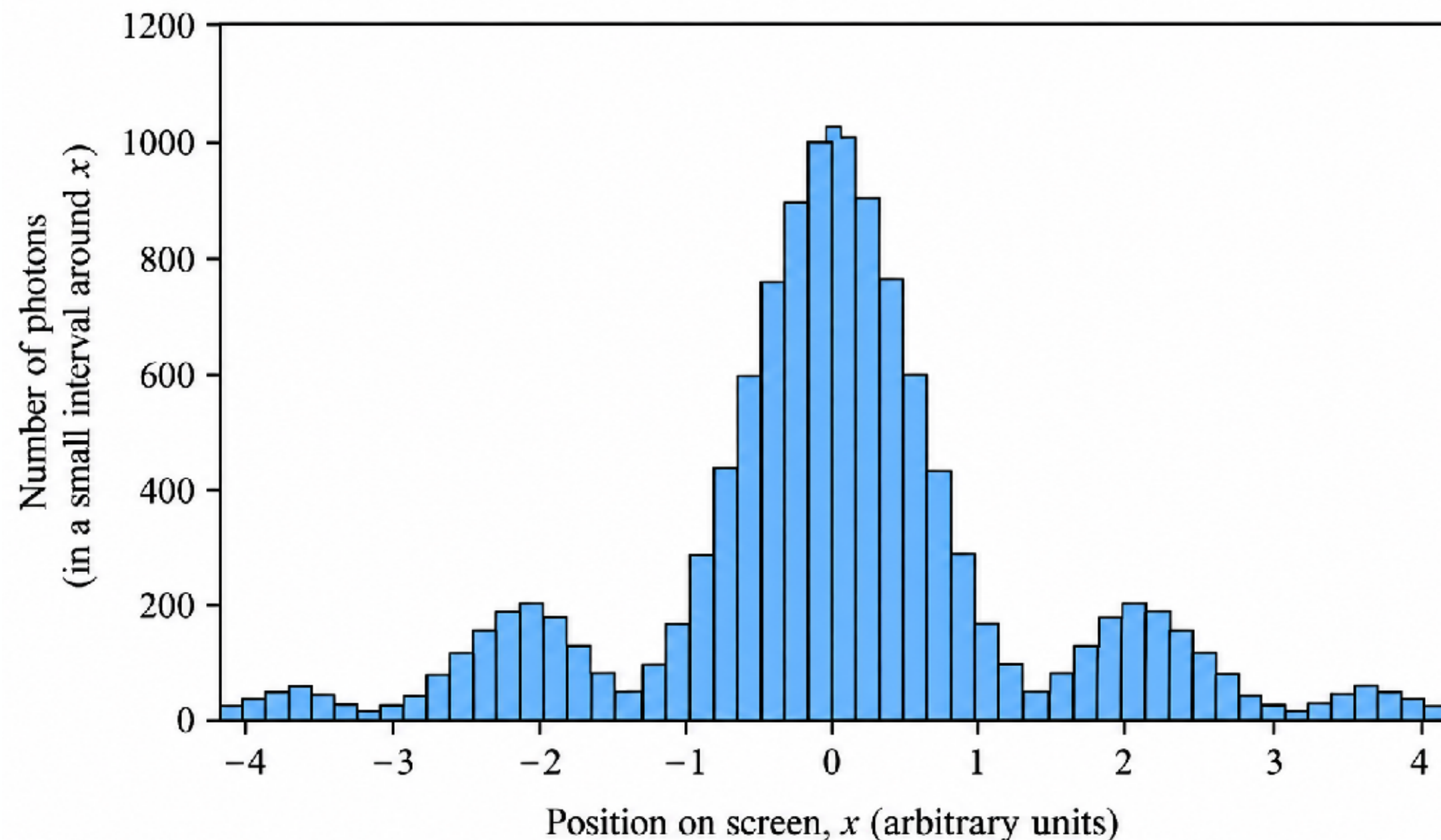
Are Intensity and No. Of photons hitting on screen connected ?

Yes, more intensity means more brightness, and more no. of photons also mean more brightness.

If the photon flux is very large, many detection events accumulate very quickly. We therefore do not have to wait long for the diffraction pattern to become visible if using a ordinary intense light source.

Diffraction in Photon picture

Distribution of detected photons (single slit)



Q: At a bright location, do we detect more particles or fewer particles?

More.

Q: If I send only **one** particle, what can “more particles here” possibly mean?

A greater chance of finding the hit here.

Diffraction can be explained by particle picture but the interpretation changes. The diffraction pattern now describes the probability of detecting a photon at different positions on the screen.

Wave - Particle Duality

- We started with an **electromagnetic wave** and then introduced its particle description in terms of **photons**.
- We found that **diffraction can be described in both pictures**:
 - Wave picture → intensity pattern.
 - Photon picture → probability of detecting photons at different positions.
- This may leave us wondering:
Are we seeing diffraction only because photons originated from an electromagnetic wave description?
- **No. Electrons also produce diffraction and interference patterns.** There is no corresponding classical electron wave that we started with.

Wave - Particle Duality

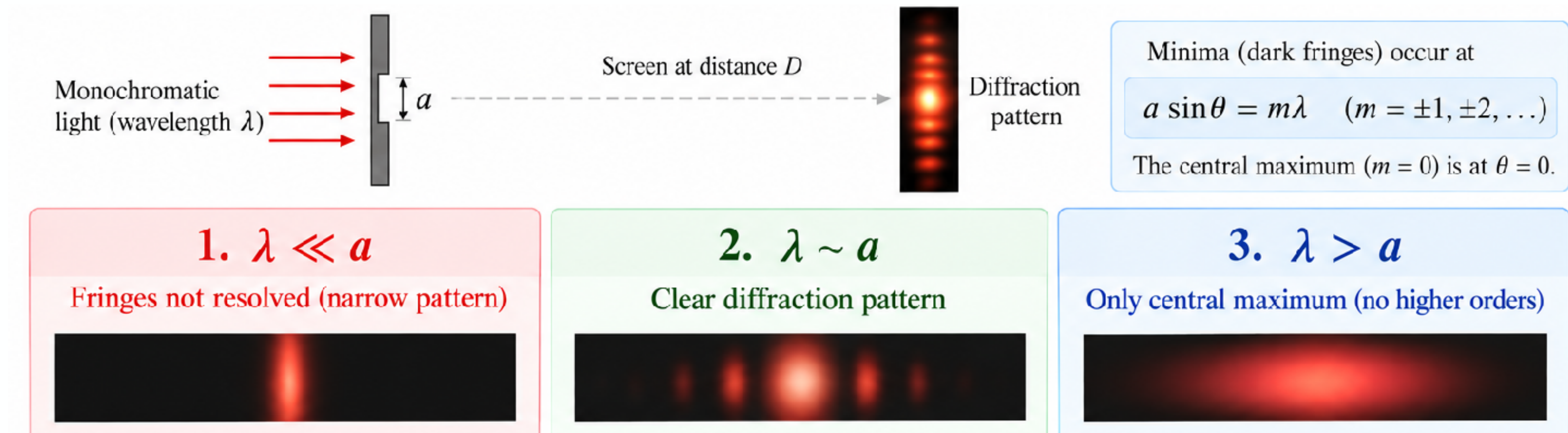
Wave-particle duality: Quantum objects can exhibit phenomena that, usually, we associate with both particles and waves.

Towards understanding the quantum objects

Q1. What does the setup tell us about the particle?

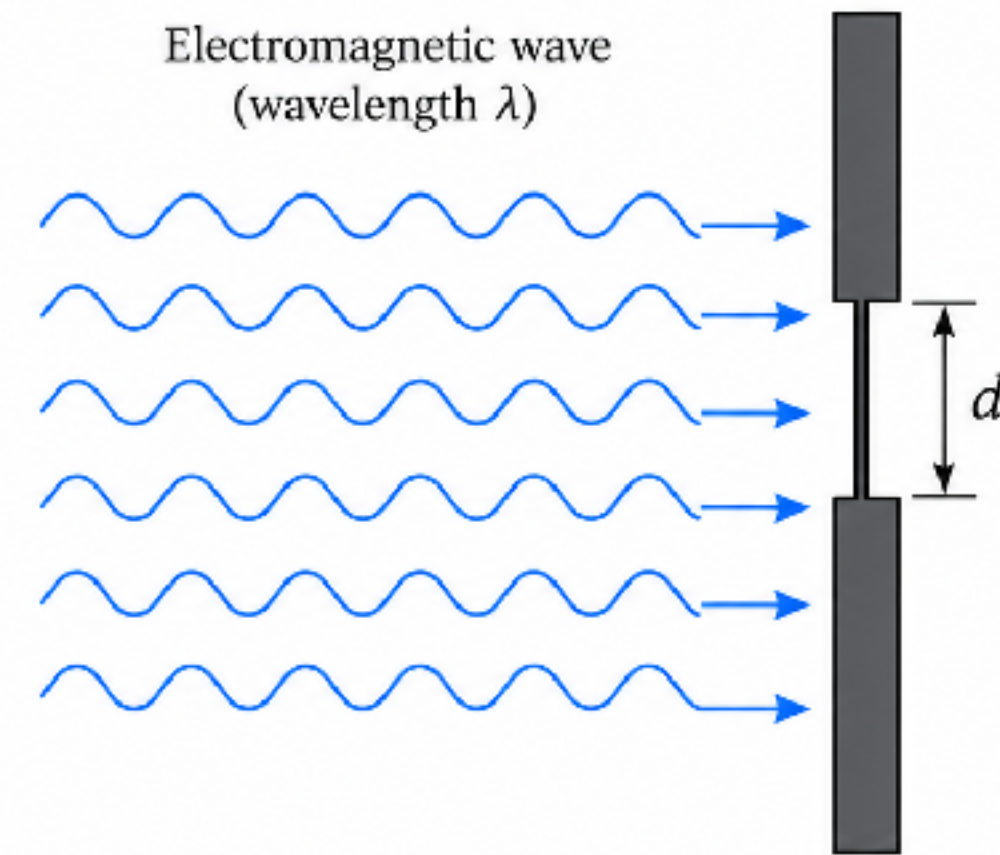
Q2. What does the diffraction pattern tell us about the particle?

The Set up

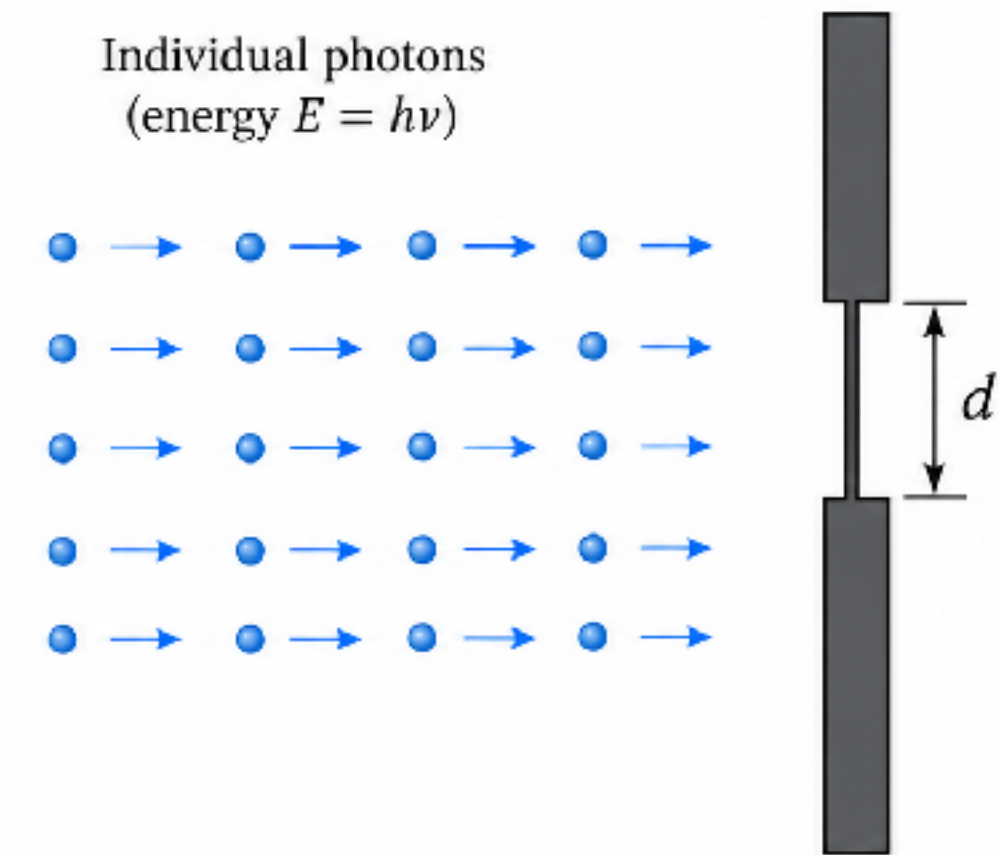


From our previous lecture, when were the diffraction fringes clearly visible? $\lambda \sim a$. For $\lambda \ll a$, diffraction still exists, but the spreading is tiny—a narrow blob; the fringes/spreading become difficult to resolve.

Wave picture



Photon picture



**The slit is present in both experiments.
But what is particle equivalent of λ ?**

Where is the crest, trough? Is particle physically oscillating up and down?

Expected for photons

For photons,

$$E = h\nu, \quad E = pc$$

According to special relativity, a massive particle cannot reach c . Since EM waves or photons propagate at c , their rest mass must be zero.

Can we obtain something resembling a wavelength using only the photon's momentum?

$$pc = h\nu = \frac{hc}{\lambda}$$

And hence,

$$\lambda = \frac{h}{p}$$

De Broglie's proposal

$$\text{Define } \lambda = \frac{h}{p}$$

- de Broglie associated a **length scale** with a particle.

Historically, this is not how I am trying to reconstruct the development. But after everything we have discussed about electromagnetic waves and photons, this relation doesn't look completely unnatural to us.

De Broglie's proposal

$$\text{Define } \lambda = \frac{h}{p}$$

- de Broglie associated a **length scale** with a particle.

Length scale for what?

The de Broglie wavelength gives a length scale that tells us when effects normally associated with waves-such as diffraction and interference-can become important for a particle.

De Broglie Wavelength

We can associate de Broglie wavelength with any object. Comparing this wavelength with the relevant length scale of the experiment tells us whether quantum aspects will be observable.

Can we (humans) diffract?

This should be fun!



Average adult:

$$m \sim 50 \text{ kg}$$

Suppose somehow:

$$v \sim 12 \text{ m s}^{-1}$$

What is your
de Broglie wavelength?

Let them calculate:

$$\lambda_{\text{dB}} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{50 \times 12} \sim 1.1 \times 10^{-36} \text{ m.}$$

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What slit width would we need for appreciable diffraction? $\sim 10^{-36} \text{ m}$

The associated quantum length scale is simply absurdly small compared with the scales on which we normally observe humans.

Wave Particle Duality

Sub-atomic particles like electron, proton, neutron can exhibit interference, diffraction...

Wave Particle Duality - Quantum Mechanics

What decides this length scale?

The particle's Momentum

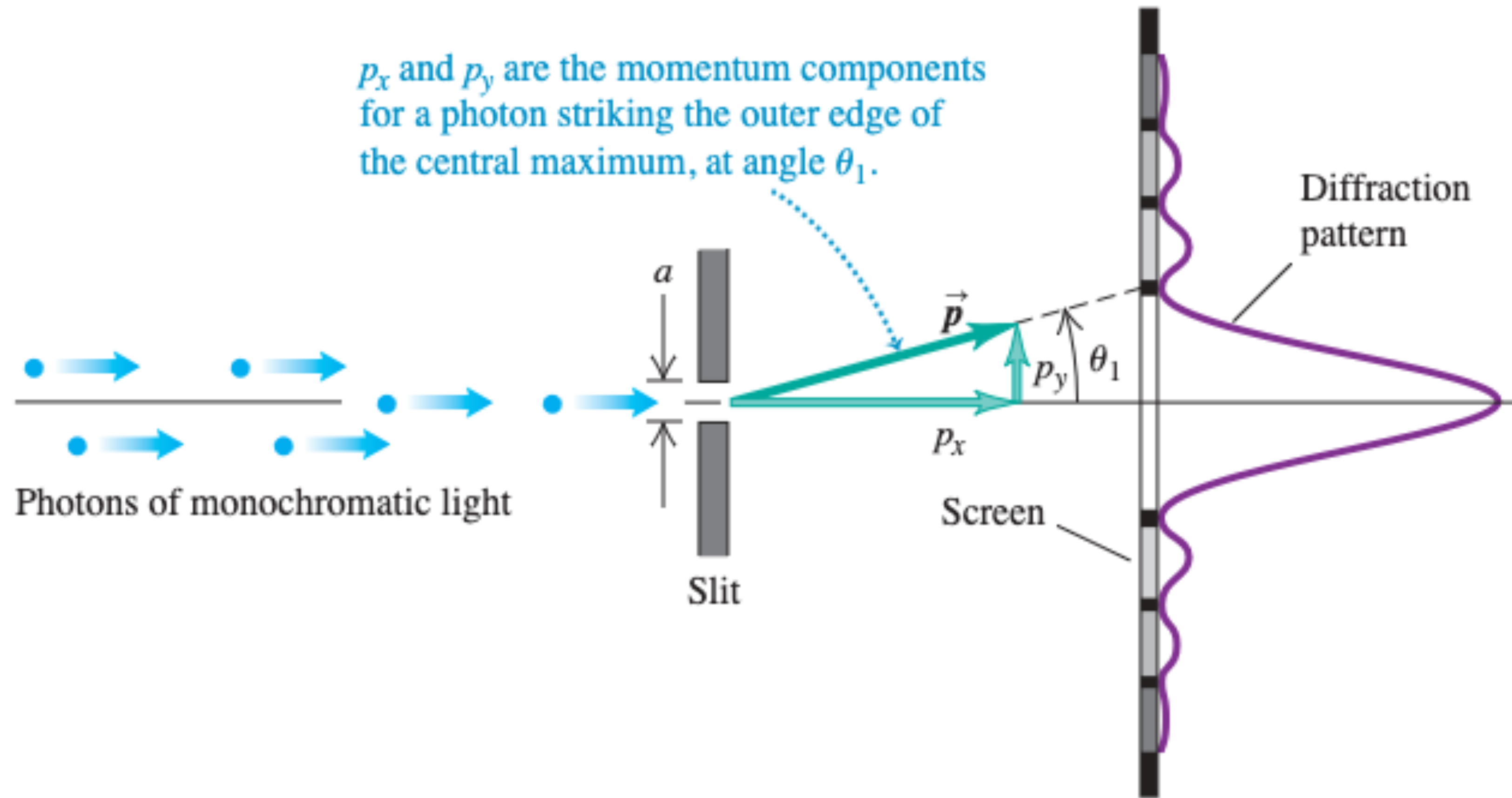
Understanding why quantities such as momentum determine these scales - and how particles acquire apparently wave-like behaviour - is part of the structure of **quantum mechanics**.

Next Question

Q1. What does the setup tell us about the particle?

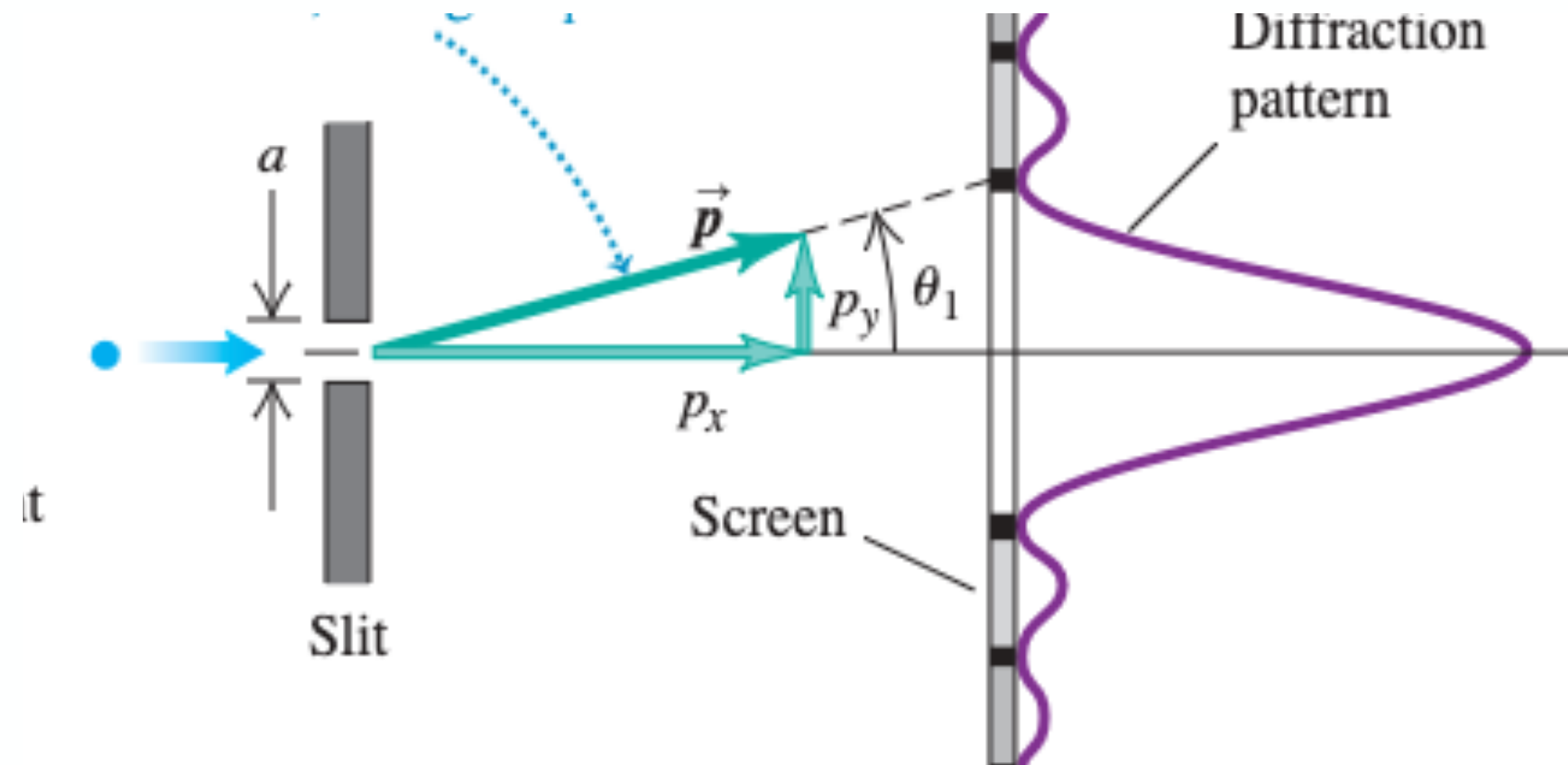
Q2. What does the diffraction pattern tell us about the particle?

Q2: Where did photon's y-momentum come from in diffraction set-up?"



Calculate p_y

Consider a photon that hit screen at the end of central maxima as shown in figure. Its momentum can be resolved in two components



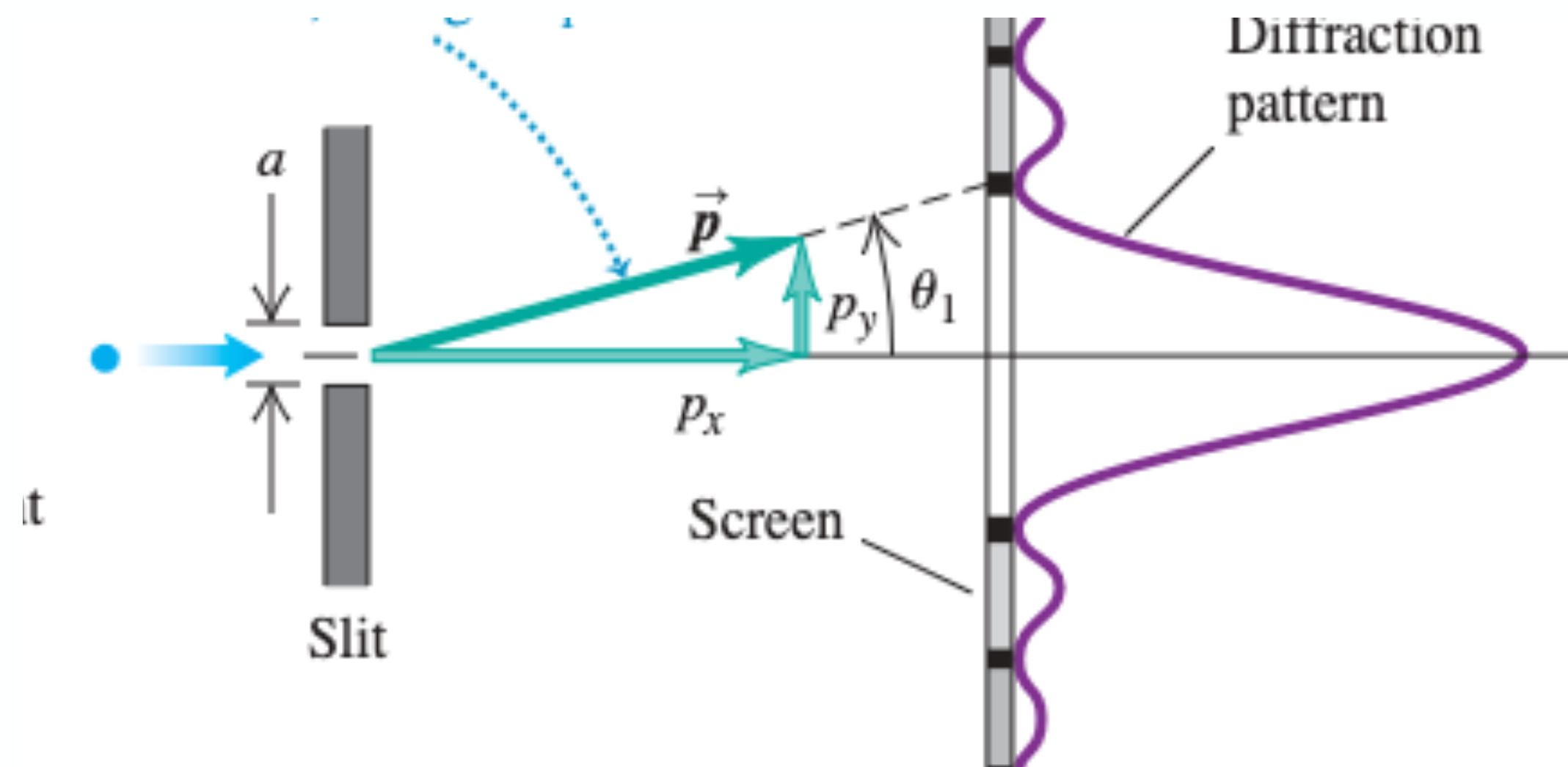
$$p_x = p \cos \theta_1$$

$$p_y = p \sin \theta_1$$

$$\frac{p_y}{p_x} = \tan \theta_1$$

Calculate p_y

For small angle, $\tan \theta_1 \sim \theta_1$



$$\theta_1 = \frac{p_y}{p_x}$$

$$\theta_1 = \frac{\lambda}{a} = \frac{p_y}{p_x}$$

$$p_y = \frac{\lambda p_x}{a}$$

Momentum of photons in central maxima

All the photons detected in central maxima have momentum in the range:

$$-\frac{p_x \lambda}{a} \lesssim p_y \lesssim \frac{p_x \lambda}{a}$$

This is spread in the y-component of momentum or the uncertainty in the p_y

Momentum of photons in central maxima

The uncertainty in the p_y can be expressed further using de Broglie's wavelength equation:

$$\Delta p_y \geq \frac{p_x \lambda}{a} = \frac{p_x}{a} \frac{h}{p_x} = \frac{h}{a}$$

Heisenberg Uncertainty Principle

If slit width is increases

$$a \uparrow \Rightarrow \theta_1 \downarrow \Rightarrow \Delta p_y \downarrow$$

A wider slit produces a narrower diffraction pattern — the photons emerge with a smaller spread in the p_y

Heisenberg Uncertainty Principle

But the slit also tells us where the photon passed:

$$\Delta y \sim a .$$

So, increased slit width implies

$$a \uparrow \Rightarrow \Delta y \uparrow , \quad \Delta p_y \downarrow .$$

Conversely,

$$a \downarrow \Rightarrow \Delta y \downarrow , \quad \Delta p_y \uparrow .$$

Better localization in position $\Rightarrow$ larger spread in momentum

Heisenberg Uncertainty Principle

Better localization in position $\implies$ larger spread in momentum

The diffraction argument suggests, in order of magnitude,

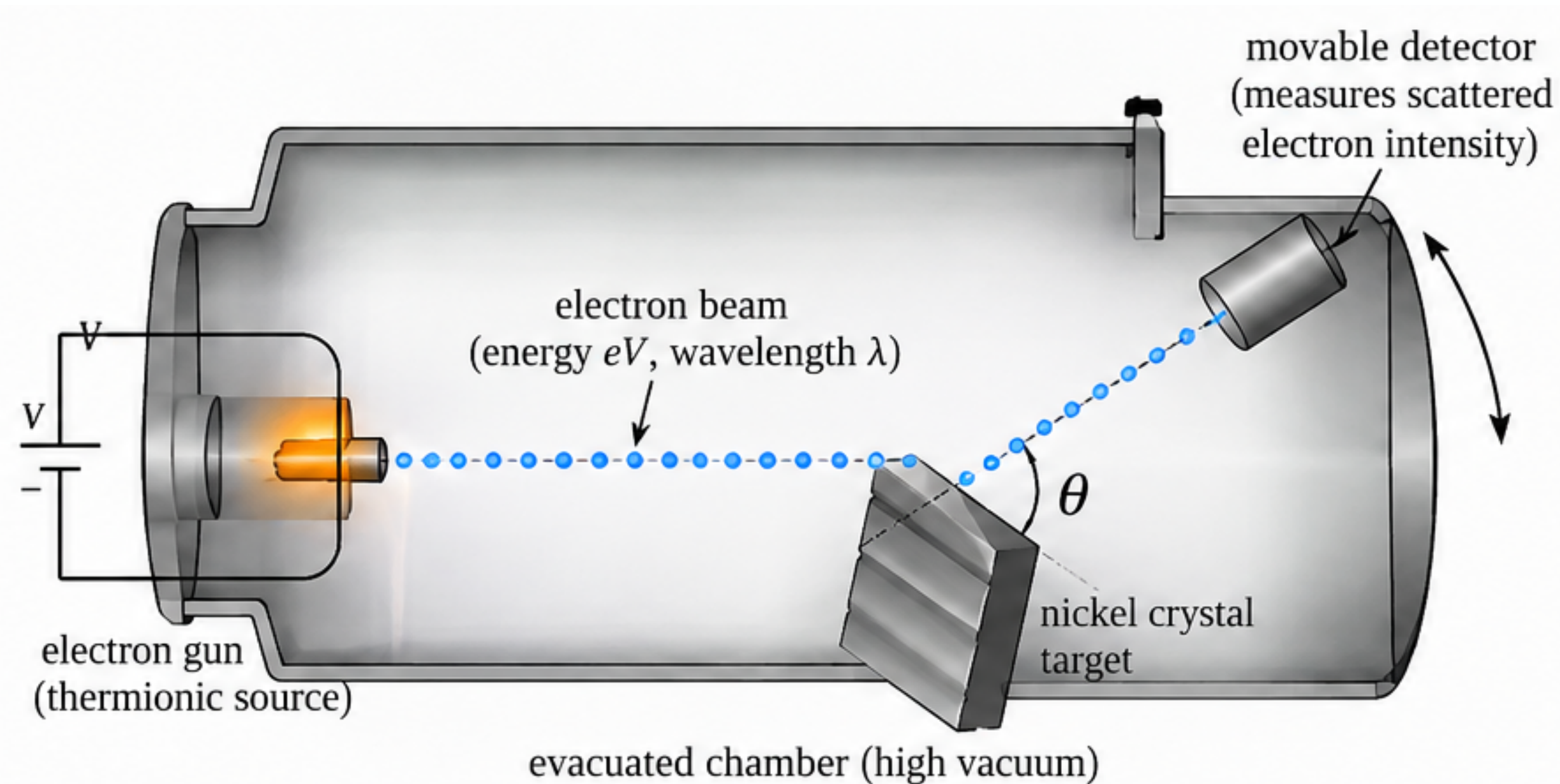
$$\Delta y \Delta p_y \sim h.$$

The precise quantum-mechanical statement is $\Delta y \Delta p_y \geq \frac{\hbar}{2} = \frac{h}{4\pi}$

Summary

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Summary - e^- diffraction



- In fact, we can observe electron diffraction experimentally — **we have such setups in our Physics Lab.**

- **For diffraction to happen, $\lambda = \frac{h}{p} \sim a$.**

- $$\lambda = \frac{h}{\sqrt{2m_e (K.E)_e}} = \frac{h}{\sqrt{2m_e eV}}$$

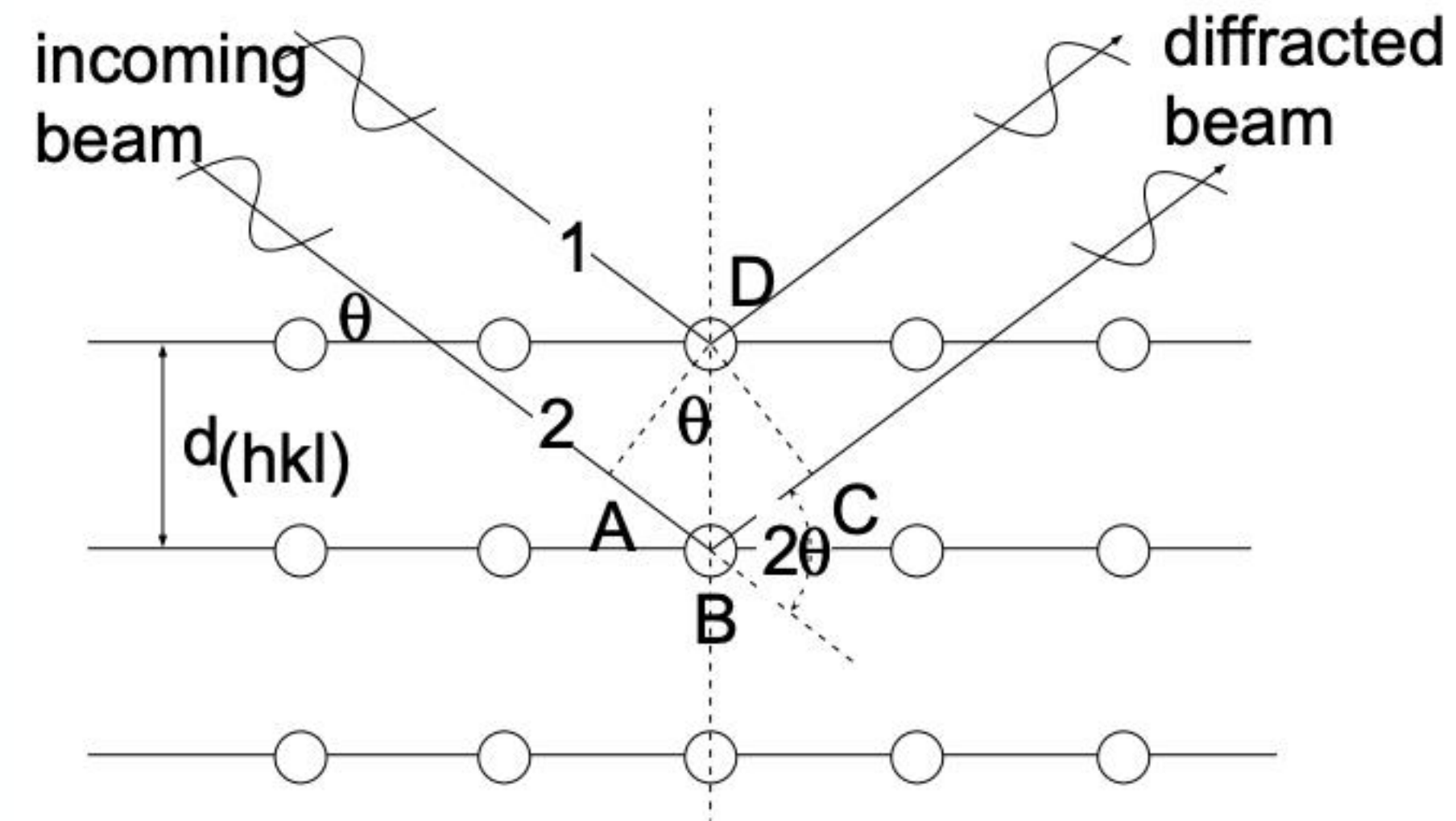
- **For 54 V power supply**

$h = 6.626 \times 10^{-34} \text{ J s}$ (Planck constant)
 $m_e = 9.11 \times 10^{-31} \text{ kg}$ (electron mass)
 $e = 1.602 \times 10^{-19} \text{ C}$ (elementary charge)
 $V = \text{accelerating voltage (volts)}$

$$\lambda = \frac{1.227}{\sqrt{54}} \text{ nm} \approx 0.167 \text{ nm}$$

For Ni (111): $d \approx 0.091 \text{ nm}$

Summary



Condition for constructive interference (Bragg's law)

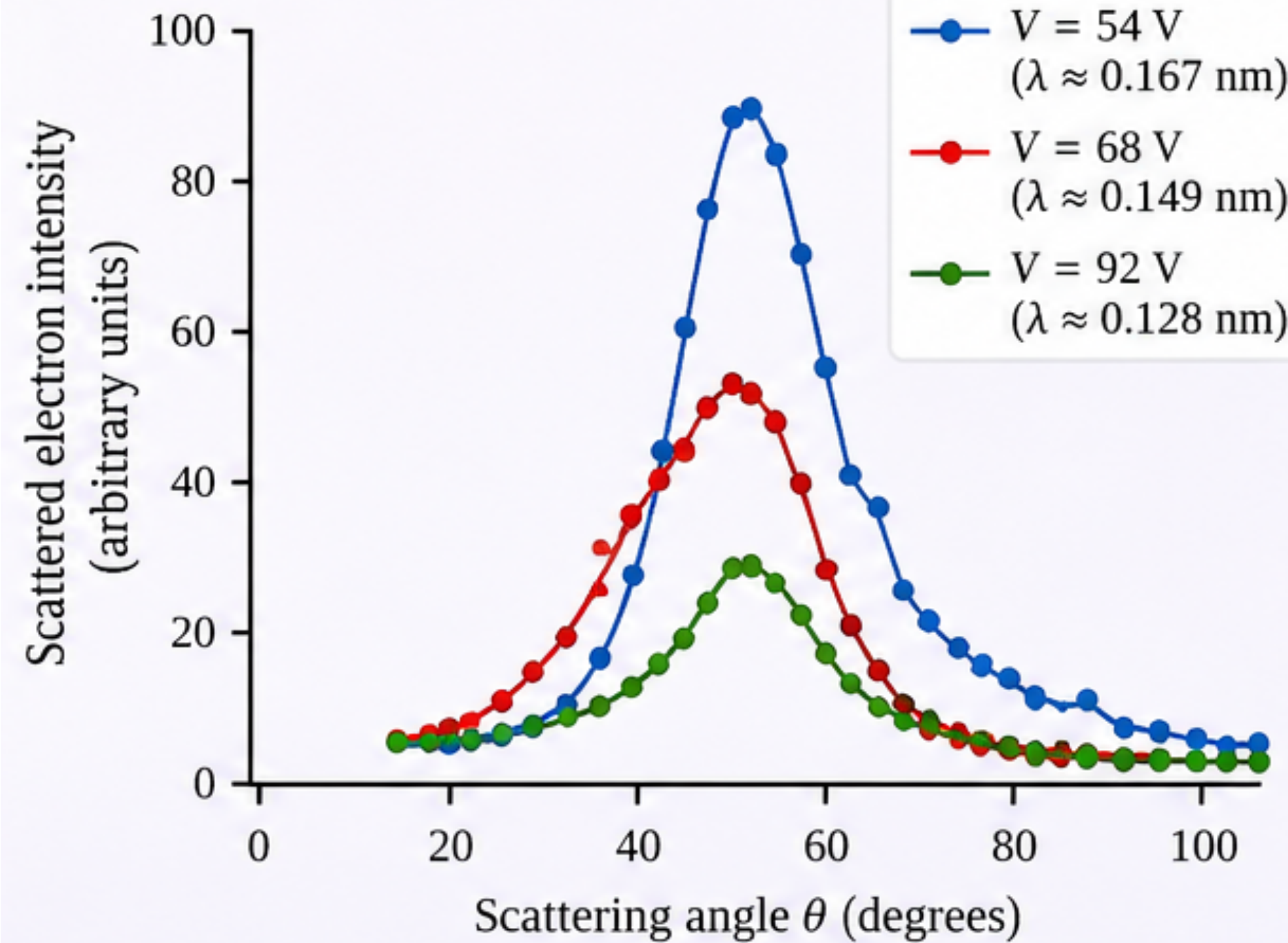
$$n\lambda = 2d \sin \theta$$

$n = 1, 2, 3, \dots$ (order)

d = interplanar spacing of crystal
(for Ni (111): $d \approx 0.091$ nm)

θ = scattering angle (from crystal plane)

Summary



Example (V = 54 V, main peak)

$$\lambda = \frac{1.227}{\sqrt{54}} \text{ nm} \approx 0.167 \text{ nm}$$

For Ni (111): $d \approx 0.091 \text{ nm}$

Using Bragg's law with $n = 1$:

$$\sin \theta = \frac{\lambda}{2d} = \frac{0.167}{2 \times 0.091} \approx 0.918$$

$\theta \approx 66.9^\circ$ (observed peak $\sim 50^\circ$ – 65° depending on crystal face, consistent with experiment)